SkyDock — Project Notes & Continuity Document

Patent & Product

- **Patent:** US20220356019A1
- **Invention:** Rooftop receptacle that resembles a skylight/solar panel and transforms into a UAV delivery landing pad
- **Package limit:** 10 lbs or less
- **Delivery method:** Drone lands OR tethers package
- **Website:** https://djt1023.wordpress.com

Project Overview

Interactive Three.js demo embedded in WordPress via iframe.
Two modes: Cinema (auto-flight) and Free Flight (WASD game).
Built with Vite + Three.js, fully offline capable.

Environment

- **OS:** Ubuntu Linux (dt@dt-OptiPlex-3020)
- **Project path:** ~/dev/projects/three-demo
- **Editor:** VS Code
- **3D tools:** FreeCAD (GLB export), Blender (GLB review)
- **Dev server:** Vite (npm run dev → localhost:5173)
- **Node/npm:** installed and working

File Structure

```
three-demo/
├── index.html              # UI, HUD, mode select, fonts (all CSS inline)
├── package.json
├── vite.config.js
├── README.md
├── NOTES.md                # this file
├── public/
│   └── assets/
│       ├── fonts/
│       │   ├── Orbitron-Regular.ttf
│       │   ├── Orbitron-Bold.ttf
│       │   ├── Orbitron-Black.ttf
│       │   ├── Exo2-Light.ttf
│       │   ├── Exo2-Regular.ttf
│       │   └── Exo2-SemiBold.ttf
│       └── (all GLB files — see below)
└── src/
    ├── main.js             # renderer, camera, animation loop
    ├── scene.js            # sky, lighting, ground, warehouse, roads
    ├── neighborhood.js     # procedural houses, trees, roads
    ├── drone.js            # drone mesh, propellers, package hook
    ├── receptacle.js       # GLB loader, 7-phase sequence controller
    └── controls.js         # cinema autopilot, game flight, delivery choreography
```

GLB Files — All 22 Unique Files

Phase 1 — Rotate dock to horizontal (10 files)

```
RotateDock_045deg.glb   ← starting position (45° matches roof angle)
RotateDock_040deg.glb
RotateDock_035deg.glb
RotateDock_030deg.glb
RotateDock_025deg.glb
RotateDock_020deg.glb
RotateDock_015deg.glb
RotateDock_010deg.glb
RotateDock_005deg.glb
RotateDock_000deg.glb   ← horizontal
```

Phase 2 — Platform extends outward (5 files)

```
RD_Plat5in.glb
RD_Plat10in.glb
RD_Plat15in.glb
RD_Plat20in.glb
RD_Plat25in.glb
```

Phase 3 — Panels slide apart (7 files) ← PACKAGE LANDS HERE

```
LSPan01.glb
LSPan02.glb
LSPan03.glb
LSPan04.glb
LSPan05.glb
LSPan06.glb
LSPan07.glb
```

Phase 4 — Panels close (reverse of phase 3, same files)

Phase 5 — Platform retracts (reverse of phase 2, same files)

Phase 6 — Platform tilts inward to urge package inside (8 files)

```
RoofDock_-05deg.glb
RoofDock_-10deg.glb
RoofDock_-15deg.glb
RoofDock_-20deg.glb
RoofDock_-25deg.glb
RoofDock_-30deg.glb
RoofDock_-35deg.glb
RoofDock_-40deg.glb     ← package urged inside here, then disappears
```

Phase 7 — Return to closed (reverse of phase 6 + reverse of phase 1)

Total unique files: 22 Phases 4, 5, 7 reuse existing files in reverse — no extra GLBs needed.

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receptacle.js — Key Architecture

```js
const ALL_PHASES = [
  PHASE1_FILES,          // index 0
  PHASE2_FILES,          // index 1
  PHASE3_FILES,          // index 2  ← completedPhase === 2 → pause + drone descend
  PHASE4_FILES,          // index 3  (reverse of phase 3)
  PHASE5_FILES,          // index 4  (reverse of phase 2)
  PHASE6_FILES,          // index 5  ← completedPhase === 5 → package disappears
  PHASE7_FILES,          // index 6  (reverse of phase 6 + phase 1)
];
```

Key methods:

- `startTransition(onPhaseComplete, onComplete)` — starts full sequence
- `pauseTransition()` — freezes after current frame
- `resumeTransition()` — continues from where paused
- `placeAt(x, y, z)` — positions receptacle in world
- `showClosed()` — jump to phase 1 frame 0
- `showOpen()` — jump to phase 3 last frame
- `setSpeed(fps)` — change animation speed (default frameDuration = 0.18s)

Preloading:

Uses `new Set(ALL_PHASES.flat())` to deduplicate — loads 22 unique files, serves phases 4/5/7 from cache. Loading bar shows progress.

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controls.js — Key Architecture

Modes:

- `'menu'` — mode select screen visible
- `'cinema'` — autopilot along CINEMA_WAYPOINTS
- `'game'` — WASD free flight
- `'delivery'` — cinematic takeover for deposit sequence (both modes)

Key constants:

```js
const GEOFENCE_CENTER = new THREE.Vector3(55, 13, -30);
const GEOFENCE_RADIUS = 15;
```

Key coordinates:

```js
hoverPoint   = new THREE.Vector3(55, 20, -22)  // observation point above house
depositPoint = new THREE.Vector3(55, 13, -30)  // on the pad surface
departPoint  = new THREE.Vector3(20, 35, 10)   // post-delivery departure
```

Key methods:

- `_startCinema()` — launches cinema mode
- `_startGame()` — launches free flight mode
- `_startDeliverySequence()` — called by BOTH cinema and game when delivery triggered
- `_cinematicMoveTo(targetPos, duration, onComplete)` — smooth drone movement
- `_triggerGameDelivery()` — full choreographed sequence with phase callbacks
- `_depositPackage(depositPoint)` — detaches package from drone, places in world
- `_setPhaseHUD(dotIdx, desc)` — updates phase dots and description
- `_showSubtitle(text)` — shows timed subtitle, auto-hides after 5s
- `_easeInOut(t)` — smooth interpolation function

Keyboard setup:

```js
// Space fires once on keydown (not polled in update loop)
// e.preventDefault() stops browser scroll on spacebar
```

Game controls:

```
W/S       forward/back       speed = 35
A/D       strafe             liftSpeed = 20
↑/↓       ascend/descend     turnSpeed = 2.5
←/→       rotate (yaw)
SPACE     trigger delivery (within geofence)
```

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neighborhood.js — Key Details

Delivery house position:

```js
export const DELIVERY_HOUSE_POS = new THREE.Vector3(60, 0, -30);
```

Delivery house fixed dimensions (to match receptacle GLB):

```js
let w = 12 + rng() * 6;
let d = 10 + rng() * 5;
let h = 6 + rng() * 4;
let roofH = 3 + rng() * 2;

if (isDelivery) w = 16;
if (isDelivery) d = 12;
if (isDelivery) h = 6;
if (isDelivery) roofH = w / 2;  // 45° — rise equals run
```

Roof geometry:

- `slopeGeo` uses 6 vertices and 8 triangles
- Angle formula: `roofH = (w/2) * Math.tan(angleDeg * Math.PI / 180)`
- At 45°: `roofH = w/2` (rise equals run)
- Gabled roof suppressed for delivery house (`if (!isDelivery)`)
- Delivery house roof comes from receptacle GLB itself

Tree variety:

```js
const h = 2 + Math.random() * 5;  // range 2-7
// foliage scales with h for proportional variety
// color randomized with HSL for pine/leafy mix
```

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scene.js — Key Details

Coordinate system (Three.js):

```
X  →  East / West
Y  →  Up / Down (height)
Z  →  North / South (negative Z = north, positive Z = south)
```

Warehouse position: `x: -160, y: 9, z: 0` **(far west)**

Delivery house: `x: 60, z: -30` **(northeast of center)**

Street runs: along X axis (east/west) and Z axis (north/south)

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receptacle.js — Placement & Rotation

In controls.js _placeReceptacle():

```js
const houseData = this.neighborhood.houses.find(h => h.isDelivery);
const ry = houseData ? houseData.roofY : 10;
this.receptacle.placeAt(
  DELIVERY_HOUSE_POS.x,
  DELIVERY_HOUSE_POS.y + ry,    // fine tune with + offset
  DELIVERY_HOUSE_POS.z
);
```

In receptacle.js _showFrame():

```js
model.position.copy(this.position);
model.rotation.y = Math.PI / 2;  // rotate 90° to face south
// model.scale.setScalar(0.0254); // uncomment if units are inches
```

Roof slant direction:

- Roof ridge runs N/S (along Z axis)
- `rotation.y = Math.PI / 2` aligns receptacle with this orientation
- Receptacle GLB has built-in 45° roof matching delivery house

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Choreography Sequence (NEXT TASK)

This is the main outstanding work. Target flow:

```
1. Drone flies from warehouse to hoverPoint (55, 20, -22)
2. Receptacle plays phases 1, 2, 3          (drone watches)
3. PAUSE after phase 3                       (pad fully open)
4. Dramatic delay (~1200ms)
5. Drone descends to depositPoint (55, 13, -30)
6. Delay (~600ms) at pad
7. _depositPackage() — detach pkg from drone, place in world
8. Drone ascends back to hoverPoint
9. Delay (~500ms)
10. receptacle.resumeTransition()            (phases 4, 5, 6)
11. After phase 6 completes                  (completedPhase === 5)
12. Remove droppedPackage from scene         (now inside roof)
13. Phase 7 plays                            (receptacle closes)
14. Drone departs to departPoint (20, 35, 10)
15. Return to menu (cinema) or game mode (free flight)
```

Key timing values to tune:

```js
1200ms  // pause after pad opens (dramatic effect)
2.5s    // descent duration to deposit point
600ms   // pause at pad before ascending
800ms   // pause before ascending
2.0s    // ascent duration back to hover
500ms   // pause at hover before collection phases
1500ms  // pause after close before departure
4.0s    // departure flight duration
```

Package detach (_depositPackage):

```js
_depositPackage(depositPoint) {
  const pkg = this.drone.userData.packageMesh;
  this.drone.remove(pkg);              // detach from drone
  pkg.position.copy(depositPoint);
  pkg.position.y += 0.35;             // sit on pad surface
  this.scene.add(pkg);                // place in world
  this.droppedPackage = pkg;          // store for later removal
}
```

Package removal after phase 6:

```js
if (completedPhase === 5) {
  if (this.droppedPackage) {
    this.scene.remove(this.droppedPackage);
    this.droppedPackage = null;
  }
}
```

Camera Notes

Blue screen issue:

- Caused by camera losing lookAt reference after drone moves away
- Fix: in update() during 'delivery' mode always lookAt receptacle position

```js
if (this.mode === 'delivery') {
  const lookTarget = this.receptacle.position.clone();
  lookTarget.y += 2;
  this.camera.lookAt(lookTarget);
}
```

Observation camera for delivery sequence:

```js
const observationCam = new THREE.Vector3(35, 25, -5);
this.camera.position.copy(observationCam);
this.camera.lookAt(this.receptacle.position);
```

Git History

```
82fb7a0  initial working build
7c4764a  fonts localized, fully offline capable   ← CURRENT HEAD
ddcae80  choreography WIP (rolled back, reachable via git reflog)
```

Useful git commands:

```bash
git log --oneline            # see commits
git reflog                   # see everything including rolled back commits
git add . && git commit -m "message"    # save checkpoint
git reset --hard <hash>      # roll back to commit
git stash                    # temporarily hide current changes
git stash pop                # restore stashed changes
```

WordPress Embed

```bash
npm run build                # creates dist/ folder
```

Upload `dist/` to web host then embed:

html

```html
<iframe
  src="https://yourdomain.com/path/dist/index.html"
  width="100%"
  height="600px"
  frameborder="0"
  allowfullscreen
  style="border-radius: 12px;">
</iframe>
```

Future Ideas (Parking Lot)

- Mobile touch controls (joystick overlay for younger users)
- Sound design — prop hum, mechanical click on receptacle deploy
- Minimap HUD element
- Day/night cycle toggle
- Second delivery house / multiple receptacles
- UAV communication protocol standard (industry standards discussion)
- SkyDock jingle — percussionist/vibraphone, 3-phase rhythmic structure matching receptacle open sequence (roll → snare build → cymbal crash)
- Particle effects (prop wash, dust on landing)
- drone.glb to replace procedural drone mesh

Key Decisions & Rationale

- **Vite** chosen over raw HTML for GLB asset serving and clean imports
- **Modular JS** (6 files) for maintainability and separation of concerns
- **Procedural neighborhood** keeps app self-contained, no external assets
- **GLB flipbook** approach (swap meshes) rather than animation tracks — matches FreeCAD export workflow perfectly
- **Fonts local** (TTF) — eliminates Google Fonts dependency for offline use
- `new Set(ALL_PHASES.flat())` deduplicates preload — phases 4/5/7 reuse cached files, only 22 loads not 37
- `previousMode` stored in _startDeliverySequence so both cinema and game return correctly after delivery completes
- `delta` **timing** throughout — frame-rate independent movement
- **Geofence** triggers delivery in game mode — cleaner than spacebar polling